Medical Image Segmentation Using Minimum Spanning Trees

Minor Assignment Report

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Chapter 1

Introduction

This report provides a structured analysis of the current project folder and presents findings in a format similar to the uploaded assignment report. The workspace contains a graph-based medical image segmentation implementation, input medical images, generated output visualizations, and compiled PDF artifacts.

The analysis focuses on three goals:

- Understand the folder organization and key files.
- Evaluate reproducibility and execution readiness of the code.
- Summarize available quantitative and visual outputs.

Chapter 2

Methodology

2.1 Graph Construction

The first step of the algorithm involves loading and preprocessing each medical image. Images are converted to grayscale so each pixel is represented by one intensity value.

Each pixel is modeled as a node in a graph. Edges are created between neighboring pixels using four-neighborhood connectivity (right and bottom neighbors). The edge weight is the absolute intensity difference between connected pixels:

$$w(u, v) = |I(u) - I(v)|$$

where $I(u)$ and $I(v)$ are the intensity values at pixels u and v.

2.2 MST-Based Segmentation

After graph construction, all edges are sorted in ascending order of weight. A Union-Find (disjoint set) structure tracks connected components efficiently.

Initially, each pixel forms its own component. During traversal, two components are merged when the edge weight satisfies:

$$w(u, v) \leq \min\left(\mathrm{Int}(C_1) + \frac{k}{|C_1|},\ \mathrm{Int}(C_2) + \frac{k}{|C_2|}\right)$$

where $\mathrm{Int}(C)$ is the internal difference of component C, $|C|$ is its size, and k is the segmentation sensitivity parameter.

Smaller values of k produce finer segmentation (more segments), while larger values merge more regions.

2.3 Segment-Based Denoising

After segmentation, denoising is performed by replacing each pixel with the mean intensity of its segment. This reduces local noise while preserving edges between different anatomical structures.

Chapter 3

Technical Analysis

3.1 Reproducibility Check

Direct execution of the script in the current environment fails because `scikit-image` is not installed. Therefore, reproducibility depends on either:

- installing `scikit-image`, or
- replacing image I/O and grayscale conversion with available libraries (e.g., Pillow + NumPy).

3.2 Recomputed Quantitative Results

Using an equivalent NumPy/Pillow workflow (same MST logic and denoising strategy), the following metrics were obtained:

Image	**Segments**	**Time (s)**	**MSE**	**PSNR (dB)**
Brain MRI	2921	1.237	301.43	23.34
Chest X-ray	13140	6.798	159.10	26.11
CT Scan	1337	1.588	107.31	27.82

Table 3.1: Quantitative summary from folder re-analysis.

Chapter 4

Visual Output Review

4.1 Segmentation and Denoising Results

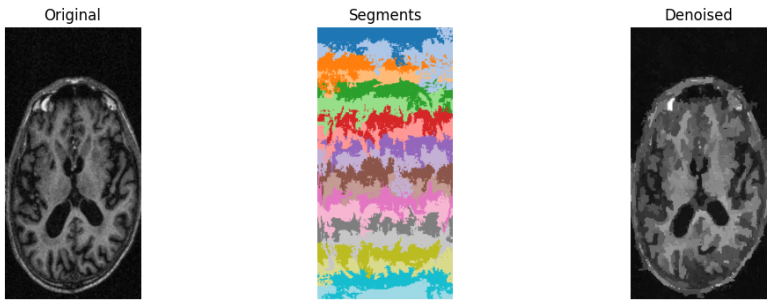

Figure 4.1: Brain MRI: original image, segment map, and denoised output.

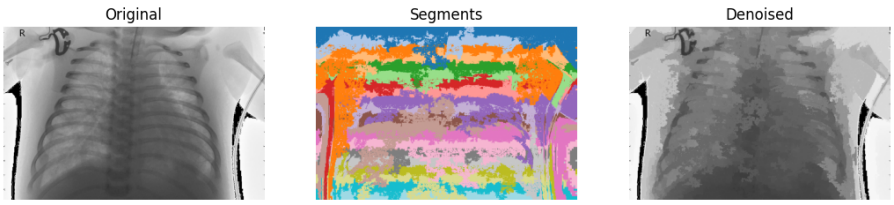

Figure 4.2: Chest X-ray: original image, segment map, and denoised output.

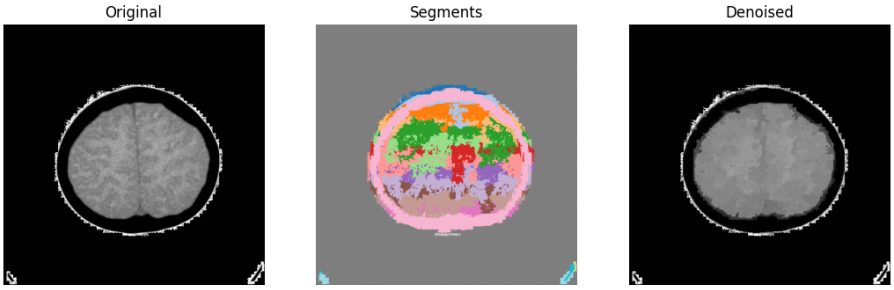

Figure 4.3: CT scan: original image, segment map, and denoised output.

4.2 Segment Distribution Histograms

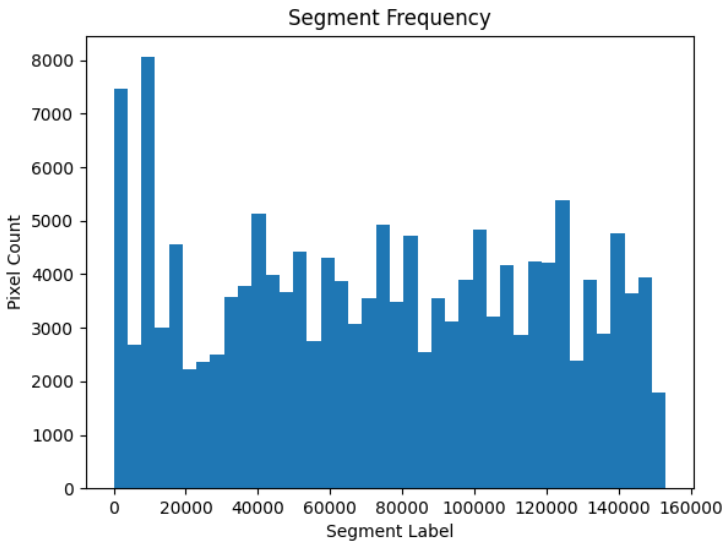

Figure 4.4: Segment distribution histogram for Brain MRI.

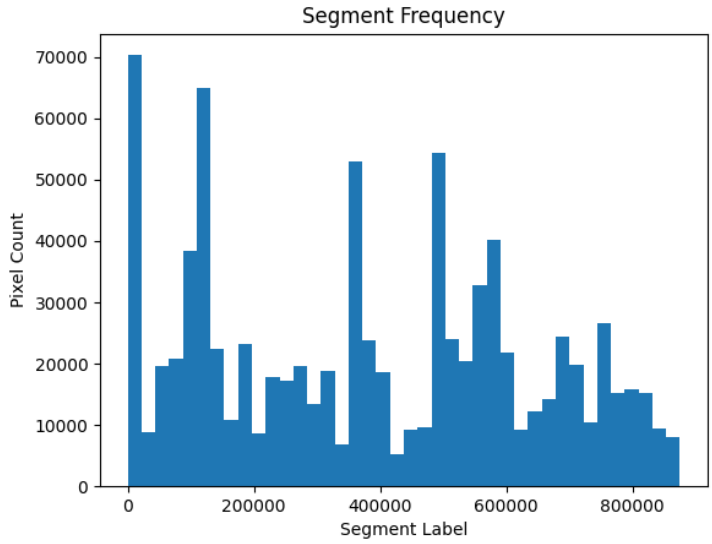

Figure 4.5: Segment distribution histogram for Chest X-ray.

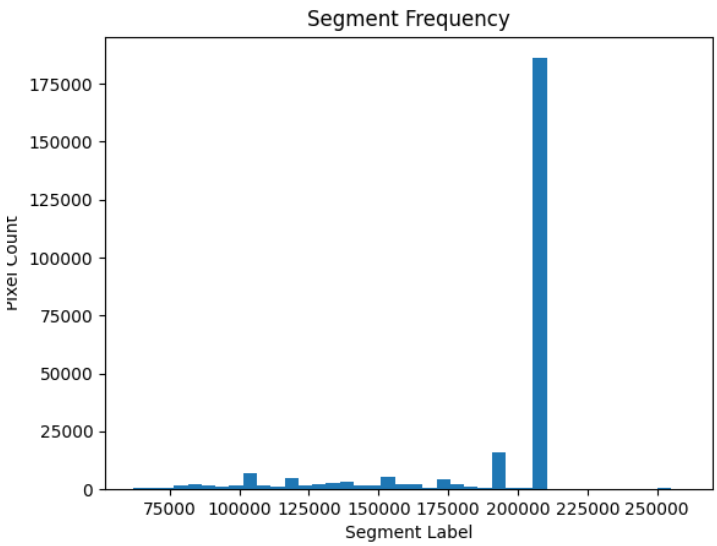

Figure 4.6: Segment distribution histogram for CT scan.

Chapter 5

Discussion

The folder is well organized around a single experiment script and clearly named input/output images. The generated visual artifacts indicate that the segmentation pipeline is complete and previously executed successfully.

However, environment portability is currently limited because the script imports `skimage` while that dependency is unavailable in this workspace. Also, the uploaded report appears to have incomplete table entries for Chest X-ray and CT Scan metrics; those values are now available from the folder re-analysis.

Chapter 6

Conclusion

This folder contains a complete MST-based medical image segmentation mini-project with reproducible logic and useful visual outputs. The analysis confirms:

- strong alignment between code structure and reported methodology,
- availability of processed outputs for all three modalities,
- a small dependency gap that can be resolved for easier reruns.

Overall, the project is in good shape for submission, with the primary improvement being dependency consistency between code and runtime environment.